Parth Mital

Vadodara, Gujarat, India | parth.mital.2004@gmail.com | linkedin.com/in/parthmital
github.com/parthmital | parthmital-portfolio.vercel.app

Education & Certifications

Vellore Institute of Technology (VIT) **2023 – 2027**
B.Tech in Computer Science & Engineering (CGPA: 8.75 / 10) *Vellore, India*

OCI 2025 Generative AI Professional **July 2026**
Oracle Certified Professional; ID: 103485063OCI25GAIOCP *Valid until July 2028*

Experience

Concept Marketing **June 2026 – July 2026**
Sole Software Developer & Maintainer (Contract) *Vadodara, India*

- Owned solo the rebuild of a 60+ employee HVAC dealership's ERP/CRM, replacing an offline single-desktop ASP.NET & SQL Server system with Next.js, TypeScript & Prisma on Vercel, **now managing 32K+ service calls**.
- Migrated **114K+ records from 42 legacy tables to Supabase PostgreSQL with zero data loss** for staff using it 6–8 hours daily, porting stored procedures to parameterised SQL builders with allowlisted columns.
- Hardened access with HMAC-signed sessions, deny-by-default role routing, nonce-based CSP, CSRF checks & row-level security; **AES-256 encrypted daily backups** & 591 Vitest tests run via GitHub Actions.

LearnLine Edustation (IntelGrader) **June 2026 – July 2026**
AI Engineer Intern *Remote*

- Won the Wellfound offer with a YOLOv8s crossout detector fine-tuned on 100 self-annotated scans; trained a multi-task MobileNetV3 quality scorer, float16 TFLite-quantised **82.5% to 2.19 MB** at 2.55 ms.
- Engineered a native Android capture app in Kotlin, Jetpack Compose, CameraX & OpenCV, bridged into the ops web portal, with TFLite page segmentation & **quad-stability auto-capture over 7 stable frames**.
- Designed concurrent question bank pipelines: PyMuPDF ingestion **extracting 1,033 questions & 383 visuals**, dual-GPU EasyOCR over 4,574 pages, & AWS Bedrock structuring with zero validation errors.
- Shaped product direction with co-founders for a Bedrock lesson page generator with **6 reusable archetypes** & html-validate, Playwright & axe-core gates; prototyped ResNet50 writer clustering & doodle lectures.

Projects & Research

The Obsidian Flask: RAG-Driven NPC Narrative Engine ***LangGraph, FastAPI, FAISS, React***

- Architected an LLM text adventure where code owns world state: an 8-node LangGraph pipeline validates NPC changes before an append-only event log, cutting attack success from 16.2% to **2.9% with 0% false rejection**.
- Evaluated 10 memory conditions over 7 worlds with vLLM; scoped FAISS memory recalled **87.7% of 20+ turn-old facts** vs 0% for a same-budget window; co-authored an IEEE-format paper (5 of 7 hypotheses supported).

NeuroScan AI: Brain Tumour Benchmark & Clinical Platform ***PyTorch, FastAPI, React, VTK.js***

- Benchmarked 24 architectures across 3 pipelines with a teammate under a faculty advisor; 5-fold ensembles of 9 backbones reached **99.42% detection accuracy** after perceptual-hash deduplication of 9,081 images.
- Compared 6 3D segmentation networks on 369 BraTS 2020 cases with Holm-corrected tests (UNet 0.8199 Dice); deployed VNet at **0.7965 Dice, 1.4M parameters & 37.5 ms**; co-authored 2 papers under faculty review.
- Built a full-stack clinical platform chaining all 3 models in a background queue (**about 4 s per scan on CPU**), with VTK.js multi-planar 3D viewing, PDF reports, TOTP 2FA, & 100 tests in CI.

Technical Skills

Languages: Python, JavaScript, SQL, Java, Bash, PowerShell
AI, ML & Data Science: Scikit-learn, Hugging Face Transformers, Sentence-Transformers, timm, ONNX Runtime, Groq, Kokoro TTS, Stable Diffusion, XGBoost, CatBoost, RapidFuzz, Pandas, NumPy
Frameworks & Web Systems: Node.js, Tailwind CSS, shadcn/ui, TanStack Query, SQLModel, SQLite, RESTful APIs
DevOps, Cloud & Tooling: Git, Linux, CI/CD, Dependabot, Semgrep, gitleaks, pytest, Vite, Label Studio
Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Software Engineering, Artificial Intelligence, Natural Language Processing, Cloud Architecture Design, Cryptography & Network Security, Digital Image Processing, Probability & Statistics